

CLASS XII · LECTURE 10

Functions, Arrays and Errors in Visual Basic

Option II · Two periods · Section C asks for arrays, loops and built-in functions

By the end of this lecture you can

- Distinguish a Function, which returns a value, from a Sub, which does not.
- Declare a one-dimensional and a two-dimensional array.
- Name syntax, run-time and logical errors with an example each.
- Use one built-in function in a small program.

Function and Sub

A **Sub** does a job and does not return a value: `Sub ShowTotal()`. A **Function** returns a value: `Function Greater(x As Integer, y As Integer) As Integer`. The multiple-choice line “a subprogram which returns a value” is a function. You call a function inside an expression. You call a Sub as a statement.

```
Function Greater(x As Integer, y As Integer) As Integer
    If x > y Then
        Greater = x
    Else
        Greater = y
    End If
End Function

Private Sub Command1_Click()
    Dim a As Integer, b As Integer
    a = Val(Text1.Text)
    b = Val(Text2.Text)
    Label1.Caption = Greater(a, b)
End Sub
```

Program 10.1 A function returns the larger integer. The result is assigned, not printed inside the function.

In VB the function's name is assigned the value to return: `Greater = x`. There is no `return x` of the C kind. Passing **by value** (`ByVal`) sends a copy, so the procedure cannot change the caller's variable. Passing **by reference** (`ByRef`, the older default) sends the variable itself, so a change inside is visible outside. The one-mark phrase “passing a copy of the variable” is pass by value.

Arrays**DEFINITION Array**

An array is a set of values of one type under one name, reached by an index. A one-dimensional array is a list. A two-dimensional array is a table, with a row index and a column index.

```
Private Sub Command1_Click()
    Dim marks(1 To 5) As Integer
    Dim i As Integer
    Dim total As Integer
    total = 0
    For i = 1 To 5
        marks(i) = Val(InputBox("Enter marks"))
        total = total + marks(i)
    Next i
    Print "Total = "; total
End Sub
```

Program 10.2 A list of five marks. The index runs from 1 to 5 because the declaration says so.

`Dim grid(1 To 3, 1 To 3) As Integer` is a three-by-three table. You need two nested For loops to visit every cell. If you write `Dim marks(5)`, VB creates indexes from 0 to 5 unless Option Base has been set, which is a trap. Writing `(1 To 5)` says exactly what you mean. Do that in the exam.

Built-in functions

A built-in function comes with the language. You call it. You do not write its body. Useful ones: `Val` (text to number), `Abs` (absolute value), `Sqr` (square root), `Len` (length of a string), `Left`, `Right`, `Mid`, `UCase`, `Int`. A long answer is: define built-in function, name three, then one program.

```
Private Sub Command1_Click()
    Dim n As Double
    n = Val(Text1.Text)
    Label1.Caption = Sqr(n)
End Sub
```

Program 10.3 `Val` and a built-in arithmetic function. `Sqr` needs a non-negative value.

Errors

Error	When you meet it	Example
Syntax	Before the program runs. The editor or compiler rejects the line	End Select missing, or a misspelt keyword
Run-time	While the program is running. It stops	Division by zero, or Val of a situation that then divides badly; a missing file
Logical	The program runs and gives a wrong answer	Average divided by 9 when ten numbers were read; using = instead of a comparison in a language that allows it

A syntax error is the cheapest to find. A logical error is the most expensive, because nothing crashes. When a question says “define an error and its types”, use this table and one example each. Three types are enough.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. A subprogram that returns a value is a:

- A) Report
- B) Function
- C) Table
- D) Label

2. Passing a copy of a variable is:

- A) Pass by value
- B) Pass by reference
- C) A syntax error
- D) A query

Short answers

S1. Define an error and its three types.

Detailed answer

L1. Explain an array and its types, with examples in Visual Basic.

L2. What is a built-in function? Explain with a program.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. B. A Sub does not return a value. A Function does.

MCQ 2. A. ByVal copies. ByRef shares the original variable.

S1. An error is a fault that stops a program or makes it give a wrong result. A syntax error breaks the grammar of the language and is caught before a successful run. A run-time error appears during the run, such as division by zero. A logical error lets the program finish with the wrong answer, such as dividing a total by the wrong count.

L1. An array stores many values of one type under one name. Each value is an element, reached by an index. A one-dimensional array is a list: Dim marks(1 To 5) As Integer creates five integers, marks(1) to marks(5). A loop For i = 1 To 5 reads them and can total them. A two-dimensional array is a table: Dim grid(1 To 3, 1 To 3) As Integer has rows and columns, visited by two nested loops. All elements have the same type. The index must stay inside the declared bounds. Arrays suit marks, names and any series where separate variables (m1, m2, m3) would be clumsy.

L2. A built-in function is supplied by Visual Basic. The programmer calls it and does not write its body. Examples are Val, which converts text to a number, Abs, Sqr, Len and UCase. Program: in Command1_Click, read n = Val(Text1.Text) and set Label1.Caption = Sqr(n). Val is required because a text box holds text. Sqr returns the square root. The function has a name, receives a value, and returns a result.